

SHETH L.U.J AND SIR M.V COLLEGE

Aim : Sensor Mote and Base Station Communication Study Develop a simple communication scenario between multiple sensor motes and a base station to study data aggregation and sink node operation

Sensor Mote and Base Station Communication

4 Sensor Motes + 1 Base Station

```
set val(chan) Channel/WirelessChannel
set val(prop) Propagation/TwoRayGround
set val(netif) Phy/WirelessPhy
set val(mac) Mac/802_11
set val(ifq) Queue/DropTail/PriQueue
set val(ll) LL
set val(ant) Antenna/OmniAntenna
```

```
set val(ifqlen) 50
set val(nn) 5
set val(rp) AODV
set val(x) 500
set val(y) 500
set val(stop) 20
```

```
set ns [new Simulator]
```

```
set tracefd [open sensor_mote.tr w]
$ns trace-all $tracefd
```

```
set namfd [open sensor_mote.nam w]
$ns namtrace-all-wireless $namfd $val(x) $val(y)
```

```
set topo [new Topography]
$topo load_flatgrid $val(x) $val(y)
```

```
create-god $val(nn)
```

```
set chan_1_ [new $val(chan)]
```

```
$ns node-config \
  -adhocRouting $val(rp) \
  -llType $val(ll) \
  -macType $val(mac) \
  -ifqType $val(ifq) \
  -ifqLen $val(ifqlen) \
  -antType $val(ant) \
```

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```
-propType $val(prop) \  
-phyType $val(netif) \  
-channel $chan_1_ \  
-topoInstance $topo \  
-agentTrace ON \  
-routerTrace ON \  
-macTrace ON \  
-movementTrace ON
```

```
# Create 5 nodes  
for {set i 0} {$i < $val(nn)} {incr i} {  
    set node_(i)[ns node]  
    node_(i) random-motion 0  
}
```

```
# Sensor mote positions  
$node_(0) set X_ 150  
$node_(0) set Y_ 250  
$node_(0) set Z_ 0
```

```
$node_(1) set X_ 250  
$node_(1) set Y_ 150  
$node_(1) set Z_ 0
```

```
$node_(2) set X_ 350  
$node_(2) set Y_ 250  
$node_(2) set Z_ 0
```

```
$node_(3) set X_ 250  
$node_(3) set Y_ 350  
$node_(3) set Z_ 0
```

```
# Base Station  
$node_(4) set X_ 250  
$node_(4) set Y_ 250  
$node_(4) set Z_ 0
```

```
# Show initial node positions  
for {set i 0} {$i < $val(nn)} {incr i} {  
    $ns initial_node_pos node_(i) 20  
}
```

```
# Base station receiver
```

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```
set sink [new Agent/Null]
$ns attach-agent $node_(4) $sink
```

```
# Sensor mote 0
set udp0 [new Agent/UDP]
$ns attach-agent $node_(0) $udp0
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 512
$cbr0 set interval_ 0.5
$cbr0 attach-agent $udp0
$ns connect $udp0 $sink
```

```
# Sensor mote 1
set udp1 [new Agent/UDP]
$ns attach-agent $node_(1) $udp1
set cbr1 [new Application/Traffic/CBR]
$cbr1 set packetSize_ 512
$cbr1 set interval_ 0.7
$cbr1 attach-agent $udp1
$ns connect $udp1 $sink
```

```
# Sensor mote 2
set udp2 [new Agent/UDP]
$ns attach-agent $node_(2) $udp2
set cbr2 [new Application/Traffic/CBR]
$cbr2 set packetSize_ 512
$cbr2 set interval_ 0.6
$cbr2 attach-agent $udp2
$ns connect $udp2 $sink
```

```
# Sensor mote 3
set udp3 [new Agent/UDP]
$ns attach-agent $node_(3) $udp3
set cbr3 [new Application/Traffic/CBR]
$cbr3 set packetSize_ 512
$cbr3 set interval_ 0.8
$cbr3 attach-agent $udp3
$ns connect $udp3 $sink
```

```
# Start packet transmission
nsat1.0"cbr0 start"
nsat1.5"cbr1 start"
nsat2.0"cbr2 start"
nsat2.5"cbr3 start"
```

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Stop transmission

```
nsat15.0"cbr0 stop"
```

```
nsat15.0"cbr1 stop"
```

```
nsat15.0"cbr2 stop"
```

```
nsat15.0"cbr3 stop"
```

Finish simulation

```
proc finish {} {  
    global ns tracefd namfd
```

```
    $ns flush-trace
```

```
    close $tracefd
```

```
    close $namfd
```

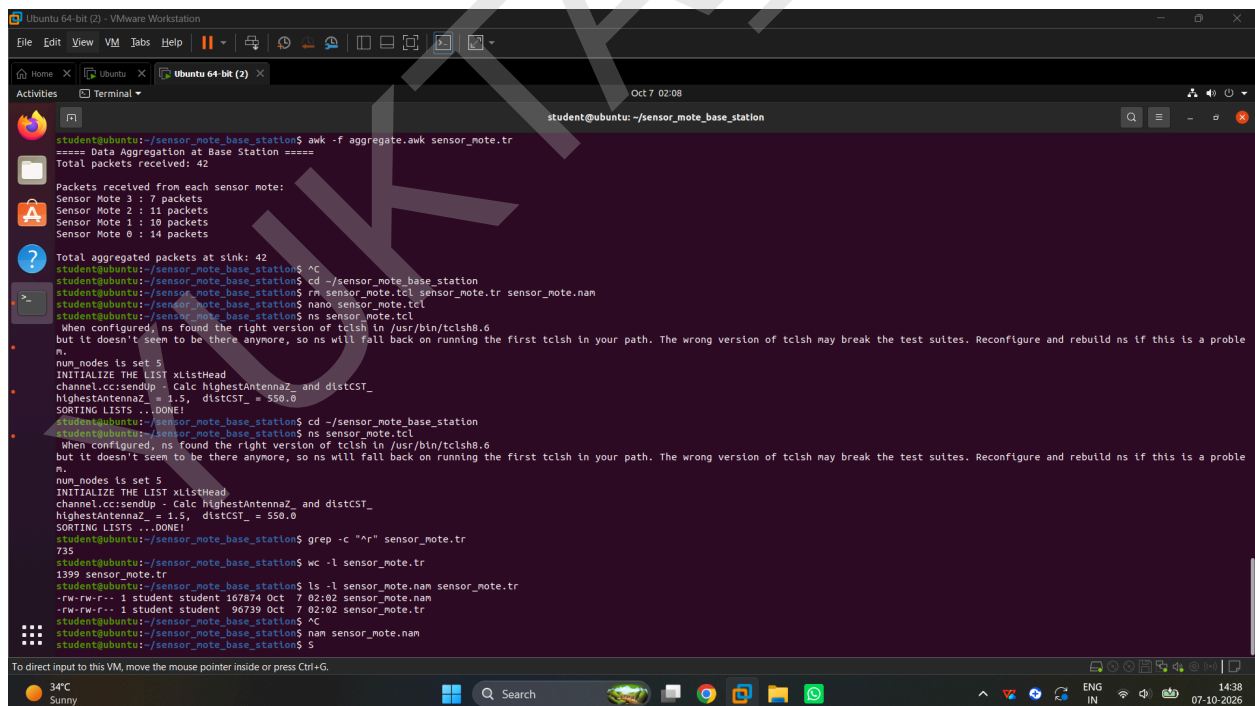
```
    exec nam sensor_mote.nam &
```

```
    exit 0
```

```
}
```

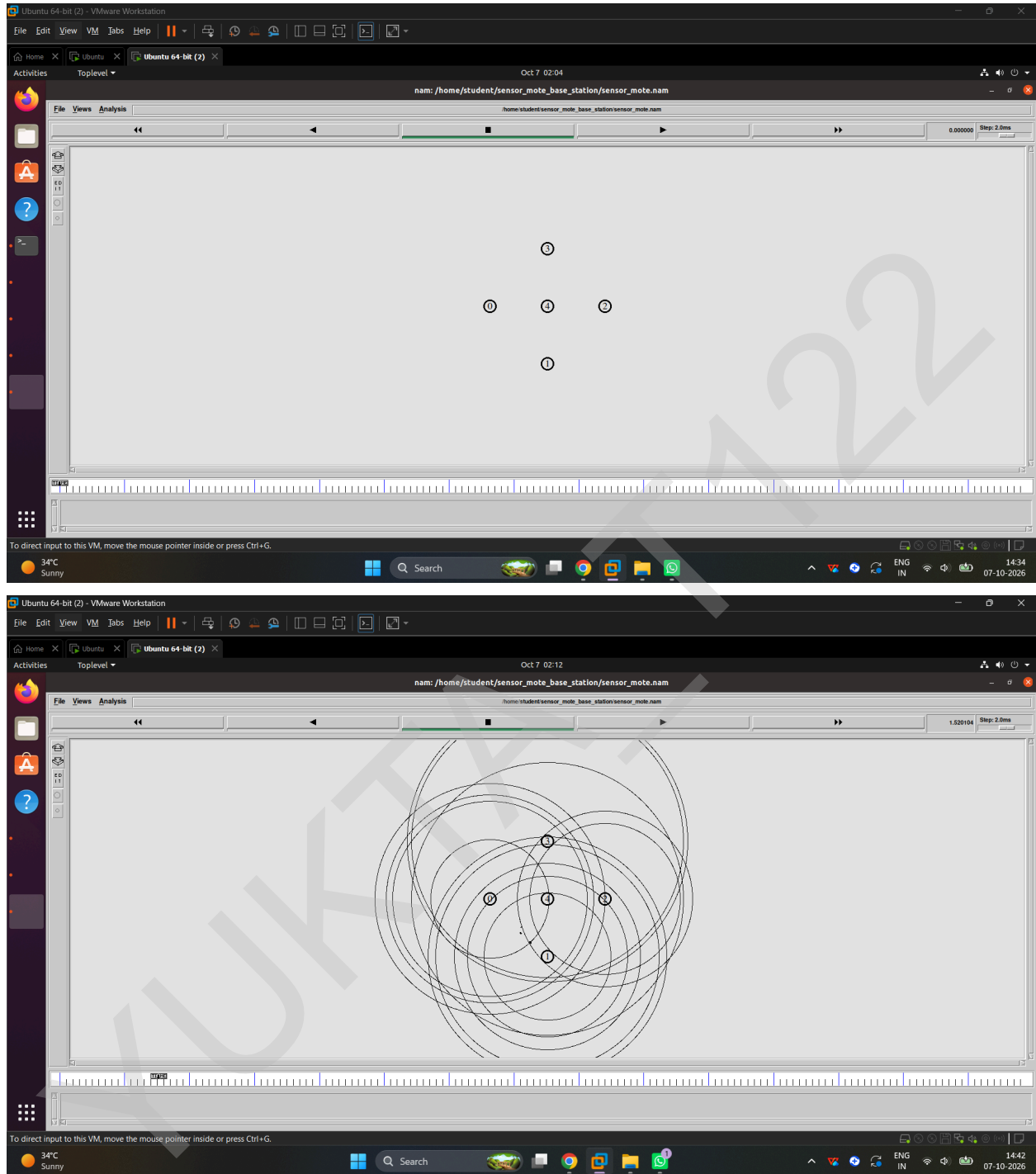
\$ns at \$val(stop) "finish"

\$ns run



```
student@ubuntu:~/sensor_mote_base_station$ awk -f aggregate.awk sensor_mote.tr  
==== Data Aggregation at Base Station ====  
Total packets received: 42  
  
Packets received from each sensor mote:  
Sensor Mote 3 : 7 packets  
Sensor Mote 2 : 11 packets  
Sensor Mote 1 : 10 packets  
Sensor Mote 0 : 14 packets  
  
Total aggregated packets at sink: 42  
student@ubuntu:~/sensor_mote_base_station$ ^C  
student@ubuntu:~/sensor_mote_base_station$ cd ~/sensor_mote_base_station  
student@ubuntu:~/sensor_mote_base_station$ rm sensor_mote.tcl sensor_mote.tr sensor_mote.nam  
student@ubuntu:~/sensor_mote_base_station$ nano sensor_mote.tcl  
student@ubuntu:~/sensor_mote_base_station$ ns sensor_mote.tcl  
When configured, ns found the right version of tcclsh in /usr/bin/tcclsh8.6  
but it doesn't seem to be there anymore, so ns will fall back on running the first tcclsh in your path. The wrong version of tcclsh may break the test suites. Reconfigure and rebuild ns if this is a problem.  
num_nodes is set 5  
INITIALIZE THE LIST xListHead  
channel.cc:sendup - Calc highestAntennaZ_ and distCST_  
highestAntennaZ_ = 1.5, distCST_ = 550.0  
SORTING LISTS ...DONE!  
student@ubuntu:~/sensor_mote_base_station$ cd ~/sensor_mote_base_station  
student@ubuntu:~/sensor_mote_base_station$ ns sensor_mote.tcl  
When configured, ns found the right version of tcclsh in /usr/bin/tcclsh8.6  
but it doesn't seem to be there anymore, so ns will fall back on running the first tcclsh in your path. The wrong version of tcclsh may break the test suites. Reconfigure and rebuild ns if this is a problem.  
num_nodes is set 5  
INITIALIZE THE LIST xListHead  
channel.cc:sendup - Calc highestAntennaZ_ and distCST_  
highestAntennaZ_ = 1.5, distCST_ = 550.0  
SORTING LISTS ...DONE!  
student@ubuntu:~/sensor_mote_base_station$ grep -c "nr" sensor_mote.tr  
735  
student@ubuntu:~/sensor_mote_base_station$ wc -l sensor_mote.tr  
1399 sensor_mote.tr  
student@ubuntu:~/sensor_mote_base_station$ ls -l sensor_mote.tr  
-rw-rw-r-- 1 student student 107074 Oct 7 02:02 sensor_mote.tr  
-rw-rw-r-- 1 student student 90739 Oct 7 02:02 sensor_mote.tr  
student@ubuntu:~/sensor_mote_base_station$ ^C  
student@ubuntu:~/sensor_mote_base_station$ nam sensor_mote.nam  
student@ubuntu:~/sensor_mote_base_station$ S
```

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